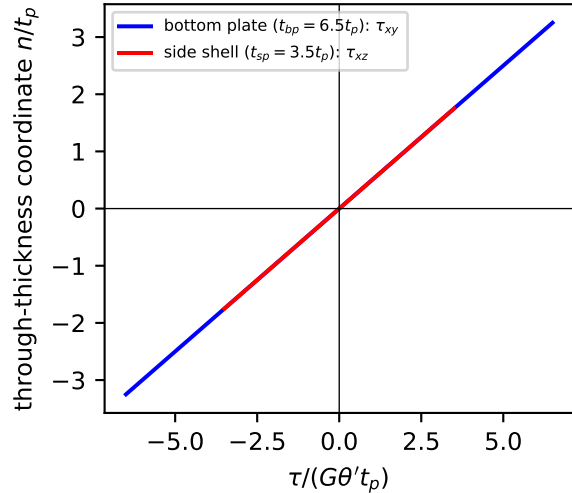
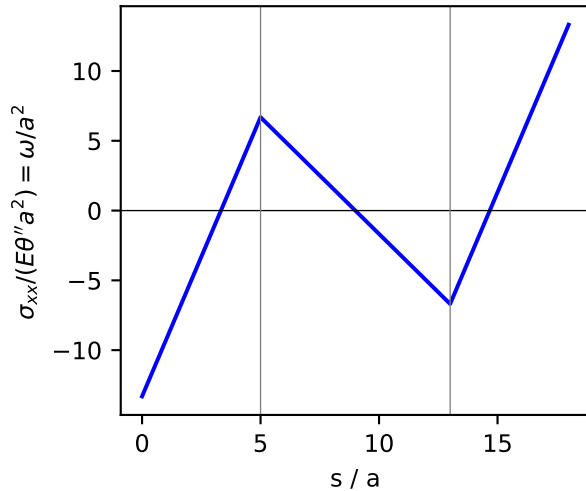


free warping (St. Venant) shear stress



constrained warping normal stress $\sigma_{xx}(s)$ (and $u(s) \propto \omega$)



constrained warping shear stress $\tau(s) = -E\theta'''S_\omega/t$

